

Tejasvini Reddy Kalluru

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SUMMARY:

OCI Certified Data Scientist with a focus on building scalable, production-ready AI systems and Ethical AI frameworks. Proven track record in eliminating algorithmic bias and implementing high-dimensional healthcare analytics with 95% variance retention. Expert in Python, SQL, and MLOps (Docker, MLflow) with specialized skills in Generative AI, RAG, and Cloud Infrastructure (AWS/Oracle) to drive measurable business impact.

WEBSITES,PORTFOLIOS,PROFILE:

linkedin.com/in/tejasvinireddykalluru

Skills:

Programming & Databases: Python (Expert), SQL (Advanced), R, NoSQL.

Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning (CNN, RNN), Generative AI (RAG, LangChain), OpenCV.

Data Engineering & MLOps: PySpark, Docker, MLflow, Git, ETL Pipelines, CI/CD.

Cloud Computing: Oracle Cloud Infrastructure (OCI Certified), AWS SageMaker, Data Governance & IAM.

Analytics & Visualization: Tableau, Power BI, Pandas, Scikit-learn, SHAP (Explainable AI).

Tools & Methodologies: Agile/Scrum, Microsoft Excel (VBA/Macros), System Hardening.

Professional /Academic projects:

1. Bias in HR Recruitment

- **Engineered an Ethical AI framework** that achieved a perfect 1.00 Disparate Impact metric, effectively eliminating gender bias while maintaining over 90% prediction accuracy.
- **Architected a bias-mitigation pipeline** using advanced feature removal and model retraining techniques, resulting in a 25% quantitative improvement in equitable hiring outcomes.
- **Ensured enterprise-level compliance** with global ethical AI standards by implementing a detection framework to identify and neutralize discriminatory patterns in automated decision-making.

2. PCA for Clinical Data Insights

- **Optimized high-dimensional medical data** by implementing Principal Component Analysis on the UCI Cleveland Heart Disease dataset, reducing 14 features into 2 critical components for faster processing.
- **Developed an end-to-end EDA pipeline** using the Python-Scikit-learn stack to automate data cleaning, feature scaling, and variance analysis for large-scale clinical datasets.
- **Uncovered diagnostic clusters** through 2D visualization, enabling the clear separation of disease vs. non-disease classes to enhance predictive healthcare insights.

3. ConversAI Chatbot (Full Stack SaaS)

- **Architected a full-stack Conversational AI SaaS** using Dialogflow and Node.js/Python, achieving high NLP accuracy and seamless real-time user interactions.
- **Implemented secure Cloud IAM role management** to govern authentication and authorization, ensuring 100% compliance during large-scale cloud deployment.
- **Integrated a real-time analytics engine** using Tableau and Power BI, enabling continuous performance monitoring and data-driven insights with zero system downtime.

4. AI/ML: Object Detection System (OpenCV)

- **Developed a real-time computer vision system** utilizing OpenCV for the automated identification and classification of objects in dynamic environments.
- **Engineered foundational AI algorithms** for advanced image processing and pattern recognition, directly applicable to industrial automation and security tasks.

- **Validated system performance** through rigorous testing of classification accuracy, demonstrating expertise in deploying applied machine learning models.

5. Data Governance & Security Dashboard

- **Designed an enterprise-level reporting dashboard** to monitor critical data security, access, and compliance metrics for organizational stakeholders.
- **Applied advanced visualization techniques** in Power BI/Tableau to translate complex security logs into actionable data governance insights.
- **Demonstrated maturity in data ethics** by building a system focused on transparency and organizational security, a primary research area for modern data-driven firms.

6. Comparative ML Evaluation (WEKA)

Engineered a rigorous machine learning workflow encompassing data preprocessing, automated feature selection, and 10-fold cross-validation to ensure maximum model reliability.

Achieved a perfect 100% classification accuracy using the J48 algorithm, significantly outperforming Naïve Bayes (96.6%) through precise hyperparameter tuning and optimization.

Validated model performance with high-precision metrics, including F1-score, ROC curves, and detailed Confusion matrices, to guarantee zero-error predictive capabilities.

CERTIFICATIONS & ACHIEVEMENTS

- **IMO (International Maths Olympiad), Gold Medal**
- Certificate in **Business Analytics** in Unified Mentor.
- Certificate in **Mobile Location Tracker** in GND SOLUTIONS PVT LTD
- Certificate in Oracle Data Platform 2025 Certified Foundations Associate
- Certificate in Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate
- Certificate in Oracle Cloud Infrastructure 2025 Certified Foundations Associate

ADDITIONAL INFORMATION

- **Languages:** Telugu(Fluent), English(Professional Proficiency), Hindi(mid level).
- **Interests:** Well Trained in Fitness, Outdoor Gameplay (Football, Shuttle, Cricket), Music
- **Hobbies :** Fitness